BEE 4750/5750

Homework 5: Gaussian Plumes and Model Validation

2026-10-14

Due Date
Thursday, 10/22/26, 9:00pm

Overview

Instructions

- Problem 1 involves evaluating the plume equation and reasoning about stability class.
- Problem 2 asks you to analyze residuals to interpret goodness-of-fit.
- Problem 3 gives you a working plume model and a set of monitor observations, and asks you to interpret the results.

This assignment is deliberately short on implementation. The work is in the interpretation.

Load Environment

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
using CSV
using DataFrames
using Statistics
using Optim
using Plots
```

Problems (Total: 50 Points)

Problem 1 (10)

The ground-level concentration downwind of an elevated point source, on the plume centreline and accounting for reflection at the ground, is

$$C(x,y) = \frac{Q}{\pi u \sigma_y(x)\sigma_z(x)} \exp\left(\frac{-y^2}{2\sigma_y(x)^2}\right) \exp\left(\frac{-H^2}{2\sigma_z(x)^2}\right).$$

A stack emits $Q = 100$ g/s at an effective height of $H = 41$ m into a wind of $u = 2.5$ m/s.

Problem 1.1 (5)

At 2 km downwind under the stability class in question, $\sigma_y = 62$ m and $\sigma_z = 23$ m.

1. Compute the ground-level concentration on the centreline at 2 km, in μg/m³.
2. Compute it again at a receptor 60 m off the centreline at the same distance. What fraction of the centreline value is it?

Problem 1.2 (5)

Stability classes run from A (very unstable, strong daytime heating) through F (stable, clear calm night). In lecture, this same stack produced a higher ground-level peak under class B (a sunny afternoon) than under class F (a clear night), much closer to the stack. The dispersion coefficients at two distances downwind are:

Distance	Class B: σ_y, σ_z (m)	Class F: σ_y, σ_z (m)
1 km	156, 110	34, 14
8 km	1001, 1064	218, 42

A school sits 1 km directly downwind of the stack, and a town 8 km downwind. The 1-hour SO_2 standard is 196 μg/m³.

1. Compute the ground-level centreline concentration at the school and at the town under each class. Which class is the worst case at each receptor?
2. The facility's consultant proposes a permit analysis that models only class B, since it produces the highest peak. What would that analysis miss? Which conditions should a permit analysis model instead, and why?

Problem 2 (5)

A dissolved-oxygen model is compared with measurements at six stations along a river, numbered in order downstream. Three versions of the model leave these residuals (model minus observation, in mg/L):

Station	1	2	3	4	5	6
Model A	+0.7	−0.5	−0.1	+0.5	−0.7	+0.1
Model B	+0.5	+0.5	+0.5	+0.5	+0.5	+0.5
Model C	−0.7	−0.5	−0.1	+0.1	+0.5	+0.7

Problem 2.1 (1)

Compute the RMSE of each model.

Problem 2.2 (3)
For each model, say in a sentence what its residuals show that the RMSE does not, and what kind of error that suggests.

Problem 2.3 (1)
A parameter that shifts every prediction up or down by the same amount (for example, a sensor's calibration offset) is changed until the fit is as good as it can be. Which model or models could that fix?

Problem 3 (35)

An industrial facility has been monitored for a night's operation. You are given twelve monitor readings in `data/receptors.csv`, and a working implementation of the plume model below.

The operator reports that conditions that night were **stability class F**, and the model below is configured accordingly. Your job is to decide whether that model is fit to be used.

```
# Pasquill-Gifford coefficients for class F. Distances in km, sigmas in metres.
class_F = (; a=34.0, c_near=14.35, d_near=0.740, f_near=-0.35,
             c_far=62.6, d_far=0.180, f_far=-48.6)

sigma_y(downwind_km, coef) = coef.a * downwind_km^0.894

function sigma_z(downwind_km, coef)
    if downwind_km < 1.0
        return coef.c_near * downwind_km^coef.d_near + coef.f_near
    else
        return coef.c_far * downwind_km^coef.d_far + coef.f_far
    end
end

function ground_concentration(downwind_km, crosswind_m, coef;
                              emission=100.0, height=41.0, wind=2.5)
    spread_y = sigma_y(downwind_km, coef)
    spread_z = sigma_z(downwind_km, coef)
    if downwind_km <= 0.05 || spread_z <= 0.0
        return 0.0
    end
    return (emission / (π * wind * spread_y * spread_z)) *
        exp(-crosswind_m^2 / (2 * spread_y^2)) *
        exp(-height^2 / (2 * spread_z^2)) * 1e6
end

monitors = CSV.read("data/receptors.csv", DataFrame)
```

Problem 3.1 (8)

Run the model at each monitor location and compute the RMSE against the observations. Report it both in µg/m³ and as a percentage of the mean observed concentration.

On that number alone, would you say the model fits?

Problem 3.2 (9)

Plot the residuals (model minus observation) against downwind distance.

1. Describe the pattern. Is it scatter, or is there a pattern?
2. What does the pattern tell you that the RMSE does not?
3. Based on the pattern alone, is the model's error more consistent with *noisy observations*, with a *wrong parameter value*, or with the *wrong model structure*? Justify your answer.

Problem 3.3 (9)

Wind speed is the most uncertain input here, and the whole concentration field scales as $1/u$. It is the obvious thing to calibrate.

Fit the wind speed by minimising RMSE against the observations, holding everything else fixed.

1. Report the fitted wind speed and the RMSE after calibration.
2. Did calibration meaningfully improve the fit?
3. Explain the result in terms of what wind speed *can* and *cannot* change about the modelled field. Your answer should refer to the residual pattern from Problem 3.2.

```
function rmse_for_wind(wind_speed)
    # TODO: run the model at every monitor with this wind speed and return
    # the RMSE against the observed concentrations.
    error("rmse_for_wind has not been written yet -- replace this line")
end
```

Problem 3.4 (9)

You are asked to certify this model as fit for use in siting a permanent compliance monitor at the facility.

1. Would you certify it? State your decision plainly.
2. The fit statistics are what they are, and you have reported them. What else do you owe the person who will rely on this model, beyond the numbers you computed?
3. Name one specific check, measurement, or piece of information you would want before certifying, and say what it would settle.
4. Suppose the operator pushes back and points out that no parameter you fitted made the model better, so the model is "as good as it can be." Respond to that argument.

References

List any external references consulted, including classmates.